

3 The Limit Comparison Test

If you attempt to use the Comparison Test but you can't establish an inequality, or your inequality "goes the wrong way", then you can use the Limit Comparison Test.

Limit Comparison Test (LCT)

Let $\sum a_n$ and $\sum b_n$ be series with positive terms, and let

$$C = \lim_{n \rightarrow \infty} \frac{a_n}{b_n}$$

- If C is any finite, nonzero number, then either both series converge or both series diverge.
- If $C = 0$, C is infinite, or the limit does not exist, then the LCT fails.

3.0.1 Example:

$$\sum_{n=1}^{\infty} \frac{1}{3^n - 2} \quad \text{Compare with} \quad \sum \frac{1}{3^n} \leftarrow \text{convergent geometric series.}$$

$$\sum \frac{1}{3^n - 2} > \sum \frac{1}{3^n} \quad \text{Converges} \quad \lim_{n \rightarrow \infty} \frac{\frac{1}{3^n}}{\frac{1}{3^n - 2}} = \lim_{n \rightarrow \infty} \frac{3^n - 2}{3^n} = 1 \leftarrow \text{finite, nonzero}$$

So both series do the same thing.

They Converge

3.0.2 Example:

$$\sum_{n=1}^{\infty} \frac{n^2 + 3n + 1}{3n^3 + 4n + 1} \quad \text{Compare with} \quad \sum \frac{n^2}{3n^3} = \frac{1}{3} \sum \frac{1}{n} \leftarrow \text{p-series. Diverges}$$

Larger numerator

$$\frac{n^2 + 3n + 1}{3n^3 + 4n + 1} ? \frac{n^2}{3n^3}$$

Larger Denominator

We Can't establish an inequality,

$$\lim_{n \rightarrow \infty} \frac{\frac{n^2 + 3n + 1}{3n^3 + 4n + 1}}{\frac{n^2}{3n^3}}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 + 3n + 1}{3n^3 + 4n + 1} \cdot \frac{3n^3}{n^2} = \lim_{n \rightarrow \infty} \frac{3n^5 + \dots}{3n^5 + \dots} = 1 \quad \text{both Diverge}$$

Series Practice Worksheet 3

featuring the comparison tests

I. For each series, circle the dominant terms in the numerator and denominator. Then determine if it converges or diverges by either using the test for divergence or comparing it with a p -series or geometric series. If you use a comparison test, tell what you're comparing to. Just get the answers here.

1. $\sum_{n=1}^{\infty} \frac{n^2 + 3}{n^3 - 2}$ $\sum \frac{n^2}{n^3} = \sum \frac{1}{n}$ $\rightarrow$ $\boxed{D}$

6. $\sum_{n=1}^{\infty} \frac{3^n - n^3}{5^n}$ $\sum \frac{3^n}{5^n} \boxed{C}$

2. $\sum_{n=1}^{\infty} \frac{4n^2 - 3n + 3}{2n^2 + n - 9}$ $\sum \frac{4n^2}{2n^2} = \sum 2$ $\boxed{D}$
 Compare with
 TFD $\lim_{n \rightarrow \infty} = 2$

7. $\sum_{n=2}^{\infty} \frac{\sqrt{n^4 + 3n}}{2n^3 + n^2 - n}$ $\sum \frac{n^2}{2n^3}$
 $\boxed{D}$ $\sum \frac{1}{2n}$

3. $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2 + n + 1}$ $\sum \frac{n^{1/2}}{n^2} = \sum \frac{1}{n^{1.5}}$ $\boxed{C}$

8. $\sum_{n=1}^{\infty} \frac{2^n + n^2}{3^n}$ $\sum \frac{2^n}{3^n} \boxed{C}$

4. $\sum_{n=1}^{\infty} \frac{\sqrt{n^2 + 11n + 11}}{n\sqrt{n}}$ $\sum \frac{n}{n^{1.5}}$ $\sum \frac{1}{n^{1/2}}$
 $\boxed{D}$ Diverge

9. $\sum_{n=2}^{\infty} \frac{\arctan n}{n^2 - n}$ $\sum \frac{\pi/2}{n^2} \boxed{C}$

5. $\sum_{n=1}^{\infty} \frac{1.01^n}{5n}$
 $\lim_{n \rightarrow \infty} \frac{1.01^n}{5n} = \infty \quad \boxed{D}$

10. $\sum_{n=1}^{\infty} \frac{5^n + n}{2^{n+4} - n^3}$ $\sum \frac{5^n}{2^{n+4}} \boxed{D}$

II. Give a detailed explanation of problems 6 and 8.